Collective-level selection from a density-dependent energy ecology

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Abstract

Evolution does not only adapt individuals; it makes new kinds of individual out of old ones. Modelling how a new level of individual first arises runs into a circularity: the standard accounts of group and kin selection begin by assuming a group and a criterion that evaluates it. Artificial life sidesteps the circularity by writing no collective-level fitness, but systems built this way have produced replicating patterns rather than populations of higher-level entities standing in relations of descent. Here I show that in a minimal model of flocking agents whose energy ecology rewards each agent for local company, collectives form, grow, divide, and compound into deep lineages, and I measure the full set of Darwinian quantities one level up: whether collectives reproduce, whether a collective-level trait is transmitted, whether reproductive success varies, and whether the heritable trait covaries with that success. It does. A selection differential on collective size is positive in every run that produces a collective lineage, survives controls for time and for division detection, and acts on a trait transmitted not through the agents' genes but through the material continuity of fission: daughters are made of their parent. The genetic channel, by contrast, shapes the collectives' internal structure yet leaves their division rate untouched. Selection at the new level therefore already operates, on an inheritance channel nobody wrote, yet produces no transgenerational trend. What this system lacks is not reproduction, heredity, variation, or their covariance, but the closure of all four into a response.

Keywords: evolutionary transitions in individuality, multilevel selection, Price equation, collective reproduction, Darwinian populations, artificial life

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1 Introduction

Evolutionary transitions in individuality, in which free-living cells became the parts of organisms and solitary insects the parts of colonies, are the reason biological complexity is layered rather than flat (Maynard Smith and Szathmáry 1995; West et al. 2015). Modelling them runs into a structural difficulty. Selection acts on entities that already reproduce differentially, so any account of how a new reproducing entity comes into being is liable to assume what it should explain. The standard formal treatments begin by specifying a fitness at the collective level, or a relatedness among a group's members, and then ask when cooperation is favoured (Okasha 2005; Hamilton 1964). Those are the right tools for asking how an existing group behaves, and the wrong ones for asking where the group came from.

Artificial life has a standard way around the circularity, which is to write no fitness at the level being studied and let differential reproduction fall out of an ecology (Ray 1991; Yaeger 1994; Channon 2001). What the field has mostly produced under that stance, however, is *pattern replication*: configurations that reproduce their own form (Langton 1984; Sayama 1999; Chan 2020). A replicating pattern is not a Darwinian population. For that one needs many higher-level entities, standing in relations of descent, differing in what they transmit and in how many descendants they leave (Godfrey-Smith 2009).

The system studied here, VitaNova, was introduced in an earlier paper (Le et al. 2025) that described its collectives qualitatively: they form, acquire a ring-like internal structure, and sometimes divide. This article supplies the measurements. Its contributions are, in order: (i) the phenomenon recurs, in that across thirty-two independent runs collectives form in thirty, and in eleven of them division events compound into lineages, the deepest running thirty-five collective generations; (ii) a collective-level trait, size at fission, is transmitted from parent to offspring under controls for replicate and for time; (iii) reproductive success varies sharply among collectives; and (iv), the measurement that motivates the title, the heritable trait *covaries* with reproductive success. The selection differential at the collective level is positive, in every run that has a lineage, under every control I could construct. Each agent's energy depends on how many others are near it, so grouping is rewarded at the level of the individual; but no rule refers to a collective, its size, its division, or its descent. The collective-level criterion that selection is using was never written, and the channel of inheritance it acts upon is not the agents' genomes but the material continuity of fission itself.

Two further measurements sharpen what this does and does not amount to. First, the genetic channel and the demographic channel come apart: agent genotype explains most of the variance in a collective's internal structure yet has no detectable effect on whether collectives divide, while size, which is inherited by fragmentation rather than by genes, is what the selection differential acts on. Second, despite positive selection and positive heredity, size at fission shows no transgenerational trend once the shared expansion of each run is controlled for: the response is absorbed. The system thus sits at a precisely measurable position on the road to a transition: a Darwinian population exists at the new level, selection already sees it, and what is missing is the closure of selection and heredity into a response, the step a full transition would have to supply.

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2 Background

2.1 What a transition requires, and what is seldom measured

Godfrey-Smith (2009) sets out three requirements for a Darwinian population: its members reproduce, offspring resemble their parents, and members differ in reproductive success. Applied one level up these become three checkable questions, and a fourth follows from Price (1970): for selection to act at the new level, the resemblance and the success must covary. The literature around these requirements is rich in mechanism: the single-cell bottleneck that purges within-collective variation (Roze and Michod 2001), fitness decoupling as the signature of a completed transition (Michod and Nedelcu 2003; Bourrat et al. 2022), fraternal versus egalitarian routes (Queller 1997), staying together versus coming together (Tarnita et al. 2013), lifetime monogamy as the enabler of eusociality (Boomsma 2009). Its central quantities, however, are usually assessed qualitatively, and almost all of it concerns what happens to a collective-level population once one exists.

Two measurement gaps compound each other. Collective-level *descent* is rarely followed: a division event and a lineage are different things, and only lineages make a population selection could act on. And collective-level *heredity* is rarely measured against the controls it needs: a parent–offspring correlation is the expected consequence of a shared temporal trend, so transmission must be separated from it by centring and permutation. The present article measures descent, heredity, fitness variation, and their covariance in one system, under one protocol.

2.2 Artificial and experimental systems

Table 1 sets the main systems against the requirements. Self-reproducing structures in cellular automata (Langton 1984; Sayama 1999; Chan 2020) replicate a configuration without any population of heritable variants; digital-evolution platforms (Ray 1991; Ofria and Wilke 2004) have genuine heredity and selection at one level only; DISHTINY (Moreno and Ofria 2019) hosts hierarchical individuality by providing explicit group-level machinery. Experimental evolution has made the most concrete progress, with snowflake yeast under settling selection (Ratcliff et al. 2012) and life cycles in bacterial mats (Hammerschmidt et al. 2014), and Black et al. (2020) state the logic: an externally imposed ecological scaffold is *sufficient* to confer Darwinian properties on collectives. The common feature of these designs is that selection at the collective level is supplied by the experimenter. The question left open is how a collective-level Darwinian population, and the selection upon it, can arise before anyone supplies either.

3 The model

3.1 Agents and controllers

The world is an 800×600 px arena with reflecting boundaries, holding 100 agents and 10 predators at initialisation. Each agent perceives others within a radius $R = 25$ px, and the separation, alignment and cohesion radii of the underlying flocking rules

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System	Collective reproduces	Collective heredity	Selection at the collective level
Self-reproducing CA (Langton 1984)	yes	no	no
Evoloop (Sayama 1999)	yes	partial	implicit
Lenia (Chan 2020)	yes	no	no
Tierra, Avida (Ray 1991; Ofria and Wilke 2004)	no	n/a	no
Polyworld (Yaeger 1994)	no	n/a	no
DISHTINY (Moreno and Ofria 2019)	yes	yes	designed in
Scaffolded collectives (Black et al. 2020)	imposed	yes	imposed
Snowflake yeast (Ratcliff et al. 2012)	yes	yes	imposed
VitaNova (this article)	**yes**	**yes**	**emerges, unwritten**

Table 1 Systems cited in discussions of artificial transitions, set against the requirements of Godfrey-Smith (2009) and Price (1970). "Imposed" marks a property supplied by the experimental protocol rather than arising from the system's own dynamics. The paper that introduced VitaNova (Le et al. 2025) reported none of the three bold entries; this article measures all three.

(Reynolds 1987) are set to the same value. Each agent carries a genome of 170 real-valued weights specifying a feed-forward network with eleven inputs, ten hidden units and six outputs. The inputs are the relative positions of three perceived neighbours, the relative position of the nearest perceived predator, the agent's own velocity, and a local density estimate. The outputs are not motions but *tendencies*: four gains in $[-1, 1]$ that weight the separation, alignment, cohesion and predator-avoidance terms, together with two outputs that set a worker or soldier state. Writing $\tau_s, \tau_a, \tau_c, \tau_p$ for the gains, $\mathbf{s}, \mathbf{a}, \mathbf{c}$ for the three flocking vectors and $\mathbf{q}$ for the displacement to the nearest perceived predator, the velocity update is

$$\mathbf{v} \leftarrow \sigma \cdot \left(\mathbf{v} + \tau_s \mathbf{s} + \tau_a \mathbf{a} + \tau_c \mathbf{c} - \max(\tau_p, 0)\, \frac{\mathbf{q}}{\|\mathbf{q}\|^2} \right), \tag{1}$$

with σ a fixed speed and the hat denoting normalisation. The flocking gains are signed, so an agent may invert any of them; the avoidance gain is rectified. An agent's controller therefore chooses a style of flocking rather than a trajectory, and what a collective looks like is a consequence of the styles its members happen to have.

3.2 Energy, reproduction, and what the rules do and do not refer to

An agent gains 1.1 energy per step and pays 1.0. On top of this it receives $+0.2$ when between two and nineteen other agents lie within 15 px, and -0.2 at twenty or more. Reproduction is asexual and occurs when energy exceeds 200; it costs 100, places the offspring nearby, and mutates its genome with rate and standard deviation 0.1. Initial energy is 50. Death is by age alone, at 1,000 steps.

The grouping subsidy is an individual-level, density-dependent payoff: it is computed for each agent from a count of its neighbours within 15 px, and it makes an

agent's reproductive success depend on having company. It is therefore a fitness consequence of grouping, and I do not claim otherwise. What the model does not contain is any term that refers to a *collective*: no rule identifies collectives, measures their size, triggers or rewards their division, or tracks their descent. Collectives, their sizes at fission, their lineages and the selection among them are reconstructed afterwards from agent positions (Section 3.3) and appear nowhere in the dynamics. The constants above have a consequence that is easier to see by multiplying them out than by running the model. An agent that never receives the grouping subsidy accumulates at most $50 + 0.1 \times 1{,}000 = 150$ energy in an entire lifetime, against a reproduction threshold of 200: it cannot reproduce under any behaviour, for any seed. Grouping in VitaNova is not an advantage that agents discover; it is an arithmetic precondition for reproducing at all. Agents also switch between worker and soldier states with a small difference in energy accounting and a predator rule that responds to groups of soldiers; ablating this caste machinery changes nothing reported here.

3.3 Collectives, lineages, and censoring

A collective is a connected component of agents at linkage distance 25 px holding at least ten members, sampled every 25 steps. Collectives are tracked between samples by membership overlap, using stable agent identifiers. A fission is one tracked collective whose members appear in two or more components at the next sample, each daughter holding at least six members and the daughters jointly retaining at least 60% of the parent. Each collective's life-history trait is its *size at its own fission*, measured when it divides and never at birth, since a daughter would resemble its parent at birth merely by being made of it. A collective's reproductive success is its number of daughters so defined. Because a collective born near the end of a run never had time to divide, fitness analyses include only collectives that divided or were born at least 2,270 steps (the 95th percentile of observed birth-to-fission waits) before the end of the run.

3.4 Implementation, validation, and statistics

All experiments use a vectorised reimplementation of the original model, 90–110 times faster; it updates agents from a per-step snapshot where the original updated them in sequence, so validation is distributional: over 78 seeds the population growth-rate ratio is 1.04, a collective divided in 32% of runs against 30%, and the size of the dividing collective does not differ (Mann–Whitney $p = 0.35$). The dataset analysed here is thirty-two runs of 12,000 steps, seeds 1–32, with a population cap of 1,500 agents. Throughout, the *run* is the statistical unit: quantities are computed within run and summarised across runs, permutation nulls shuffle only within run, and confidence intervals are bootstrap intervals over runs.

4 Results

4.1 The phenomenon, and how often it recurs

Figure 1 shows the arc of the phenomenon in two independent runs: dispersed agents aggregate; the aggregate acquires the ring-like internal structure quantified by the

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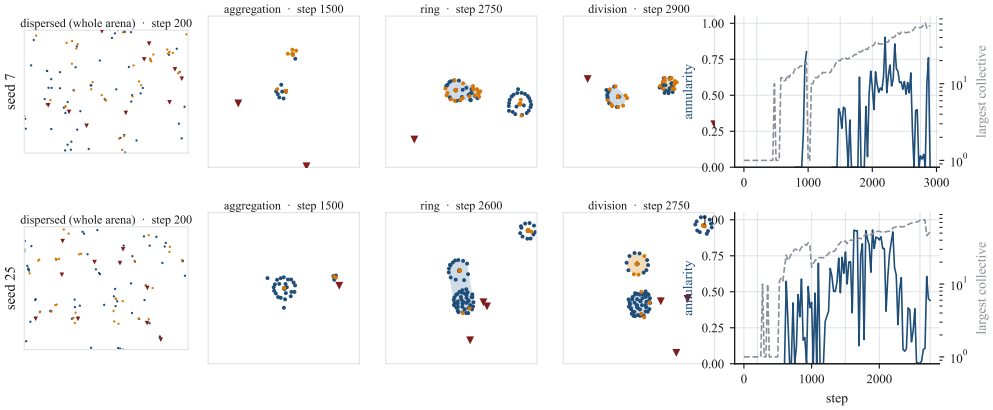


Fig. 1 The phenomenon, twice. Two independent runs (seeds 7 and 25; rows). Columns 1–4: the dispersed population early in the run (whole arena), then aggregation, ring formation and division, zoomed on the focal collective; workers blue, soldiers amber, predators dark red. In the division panels the two daughters, identified by membership overlap with the parent, are shaded separately. Column 5: annularity of the largest collective (solid, left axis; shown when it holds at least 15 members) and its size (dashed, right axis, logarithmic), with the four snapshot moments marked.

annularity statistic; the collective grows; and it divides into daughters that go on to repeat the cycle. Across the thirty-two runs, collectives of at least ten members formed in thirty; at least one division occurred in eleven; those eleven runs produced 402 division events in all, and in every one of them the divisions compounded into a lineage at least three collective generations deep, the deepest reaching **thirty-five generations**. Division is reproduction rather than fragmentation: comparing each daughter at *its own* fission with its parent at *its* fission, the median size ratio is 0.99 (IQR 0.79–1.14, $n = 382$ pairs). Daughters are not debris: they grow to the parental condition and divide in turn.

4.2 A collective-level trait is transmitted

Across the 382 parent–daughter pairs in which both eventually divided, the regression of daughter size at fission on parent size at fission has raw slope +0.85 ($r = +0.86$); centred within run the slope is +0.73, against a within-run permutation null of zero (two-sided $p < 0.0005$); additionally residualising both sizes on the daughter's fission step, the control that removes a shared time trend, leaves it at +0.72 (Figure 2b). Size at fission is therefore a transmitted collective-level trait, not an echo of replicate identity or of when in the run the pair happened to live.

4.3 Reproductive success varies sharply

Of the 873 collectives scoreable for fitness under the censoring rule, 471 left no daughters, 390 left two, eleven left three and one left five (Figure 3a). The opportunity for selection, the variance of relative fitness, is $I = 1.20$. Most collectives are evolutionary dead ends; a large minority double themselves.

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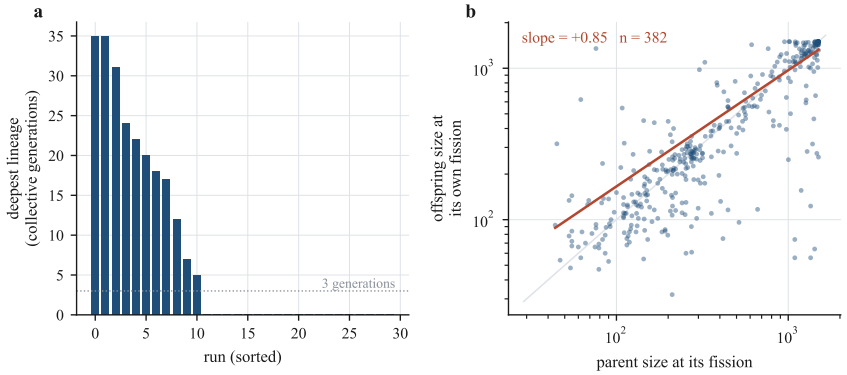


Fig. 2 The collective-level population. (a) Deepest lineage per run, sorted, over the thirty-two runs; eleven runs exceed three collective generations and the deepest reaches thirty-five. (b) Heredity: each daughter's size at its own fission against its parent's size at its fission ($n = 382$ pairs, log axes); the grey line is identity, the red line the fitted regression.

4.4 The heritable trait covaries with success

The quantity the previous three results leave open is the one that decides whether this population is merely *visible to Darwin* or already *visible to selection*: the covariance between the transmitted trait and reproductive success. Measured as a standardised selection differential within each run (the covariance between size, z-scored within run, and relative fitness), selection on size at birth is $S' = +0.57$ (bootstrap 95% CI +0.48 to +0.66) and is positive in each of the eleven runs with a lineage; on peak size it is +0.61 (Figure 3b). Residualising both trait and fitness on birth step within run leaves $S' = +0.57$: the differential is not a shared time trend.

Two artefacts could manufacture that covariance, and both are excluded. First, the fission detector requires daughters of at least six members, so very small collectives cannot *register* a division; restricting the analysis to collectives born well above the detection floor (birth size ≥ 15, ≥ 20, ≥ 25) leaves the trait–success relation unchanged in each case. Second, and decisively, the covariance *compounds* down lineages rather than living only in cross-section: a parent's size at fission predicts its daughters' own reproductive success (within-run $r = +0.18$, permutation $p < 0.0002$, $n = 704$ daughters), and does so after residualising on the daughter's birth step ($p = 0.0004$; Figure 3c). Heredity times selection is running across generations, not merely within a snapshot. Fitness itself, by contrast, is not heritable: among dividing parents, leaving two versus three daughters does not predict the daughters' own fitness ($p = 0.82$). This is expected, since nearly all dividers leave exactly two.

4.5 No response: the trend selection should produce is absorbed

Positive heredity (+0.72) and a positive selection differential (+0.57) predict, by the breeder's-equation logic, a rise in size at fission along lineages. The raw data obliges: median size at fission climbs from 88 at generation zero to 426 by generation eight

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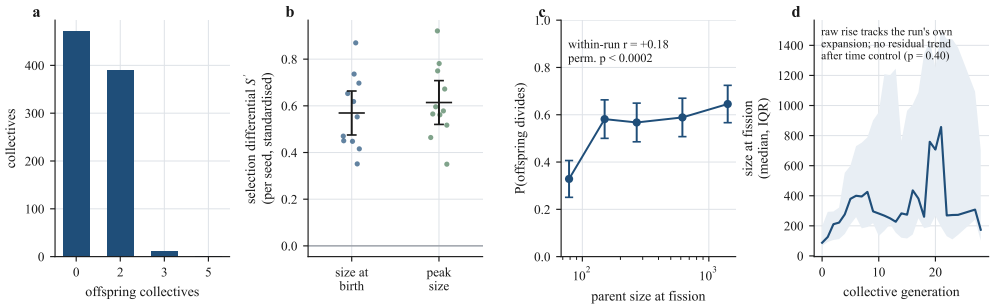


Fig. 3 Selection at the collective level. (a) Offspring-collective numbers over the 873 scoreable collectives. (b) Standardised selection differential S' on size at birth and peak size: one dot per run, mean and 95% CI across runs. (c) Probability that a daughter collective itself divides, against its parent's size at fission (quintiles, Wald 95% CIs); the within-run correlation and permutation p are printed. (d) Size at fission along the lineage (median and IQR by collective generation): the raw rise tracks each run's own demographic expansion, and no residual generational trend survives a within-run time control.

(+16.9 members per generation, within-run $r = +0.33$). The obligation is illusory, however. Generation depth and elapsed time are nearly collinear, since deep generations can only occur late, and once both size and depth are residualised on the fission step within run, no generational trend remains (+3.6 members per generation, $r = +0.04$, permutation $p = 0.40$; Figure 3d). The honest statement is that the observed rise is indistinguishable from each run's overall demographic expansion, and the design cannot fully separate the two. What can be said is that selection and heredity, each individually robust, do not visibly compound into directional change over thirty-five generations. This is the signature situation of evolutionary stasis despite selection and heritability, familiar from natural populations (Merilä et al. 2001), here reproduced in a system where every variable is observable.

Three mechanisms could absorb the response, and the data cannot yet decide among them. Reproduction by fragmentation *resets* the trait each generation: a daughter inherits her parent's scale only through the material she is made of, and must regrow before dividing. The ecology is density-dependent: the same crowding term that subsidises grouping penalises it past twenty neighbours, taxing exactly what selection favours. And part of the covariance may be environmentally induced rather than causal: conditions that enlarge collectives, such as a food-rich, crowded phase of the run, may independently promote division, the classic confound of Merilä et al. (2001) and the reason measured selection in the wild so often fails to produce the predicted response.

4.6 Two channels of inheritance, only one visible to selection

The earlier batch of experiments on this system (reported in the public repository and summarised here) locates what the population depends on and what evolution contributes, and the combination is the most theoretically pointed result of the system. The population needs its ecology: with the grouping subsidy off, no agent can ever

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reproduce (the arithmetic of Section 3.2) and no collective exceeds twenty-one members. It also needs unbounded growth: adding a crowding cost that *stabilises* the ring structure collapses the deepest lineage from nineteen generations to one, with three of ten populations going extinct. Individual-level stability and lineage-level depth trade off directly.

Evolution, meanwhile, shapes the collectives but does not make them reproduce. Agent genotype explains 82.9% of the variance in collective structure against 1.7% for initial conditions (twelve genotypes crossed with eight initial conditions, $F = 37.1$, $p < 0.0001$); selection on agents raises median peak annularity from 0.59 (clonal) to 0.92 (evolving, $p = 0.0002$); and the rate of collective division is unchanged between clonal and evolving populations (28/40 versus 30/40 runs, Fisher $p = 0.80$). Set beside Figure 3, this cleanly separates two channels of inheritance running through the same system. The *genetic* channel, mutation of the agents' genomes, carries the collectives' internal structure and is invisible to collective-level selection, since it does not touch division. The *material* channel carries size: daughters are made of their parents, the "material overlap" that Griesemer (2000) places at the centre of reproduction. It is this channel, which nobody wrote and no genome encodes, that the measured selection differential acts upon.

5 Discussion

5.1 Where this sits

By the requirements of Godfrey-Smith (2009), VitaNova's collectives are a Darwinian population: they reproduce (402 divisions compounding to thirty-five generations), daughters resemble parents (slope +0.72 under controls), and success varies ($I = 1.20$). By the further requirement of Price (1970), they are moreover already under selection: the transmitted trait covaries with success in every run with a lineage, robustly to time and detection controls, and the covariance propagates down lineages. None of this was specified at the level where it is measured. The model rewards individuals for local company, but contains no collective-level criterion for these results to be an artefact of, which is precisely what makes the measurement worth making: the population, its heredity, and the selection upon it are all *outputs* here, where in scaffolded experimental systems (Black et al. 2020; Ratcliff et al. 2012) selection is an input. What the system still lacks is equally precise: a response. Selection and heredity do not compound into directional change; the Darwinian machine at the new level is assembled and running, and the vehicle does not move.

It is worth listing what was *not* required to reach this point: no relatedness mechanism beyond what fission incidentally produces, no single-cell bottleneck (Roze and Michod 2001), no division of labour (the caste machinery ablates away), no fitness decoupling (Michod and Nedelcu 2003; Bourrat et al. 2022). Two things were required, and both are ecological rather than genetic: an energy ecology in which grouping is an arithmetic precondition for reproducing at all, and room for unbounded growth, without which lineages collapse to a single generation even as individual collectives become *more* stable.

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5.2 The two-channel structure, and social niche construction

The dissociation between the genetic and material channels gives a concrete, measured instance of a structure that transition theory has mostly had to posit. The agents' genes build the collective (82.9% of structural variance), so the group phenotype is overwhelmingly a genetic product; yet the group-level Darwinian process runs on a different inheritance medium entirely, the physical handover of members at fission. In Griesemer (2000)'s terms, reproduction at the new level is constituted by material overlap before it is served by any replicator. The arrangement is also a measurable case of social niche construction (Ryan et al. 2016): the controllers that evolution tunes are tuning, in turn, the group structure that determines which agents receive the energy subsidy: agents construct the collective environment that selects agents. What the present measurements add to that picture is the demonstration that this loop can assemble a second, selectable Darwinian level without the loop itself being selected for doing so.

5.3 What a transition would still need

Framed in the learning terms of Watson and Szathmáry (2016), the system has acquired a memory (collective heredity) and an error signal (a positive selection differential), and still fails to learn: the response is absorbed by trait-resetting at fission, by density dependence, or by environmental covariance; the data cannot yet say which. This localises the "hard part" of a transition differently from the usual accounts. The difficulty here was not getting collectives to reproduce, nor getting them to inherit, nor getting selection to see them; all three fell out of a sufficiently generative ecology. The barrier is the closure of these into cumulative change, the step that would make collective-level adaptation, rather than collective-level selection, the right description. The natural next experiments follow directly: impose a collective-level scaffold (Black et al. 2020) on the spontaneous population and measure how much imposed selection is needed before a response appears; and intervene on the three absorption mechanisms separately (fission fraction, crowding schedule, resource homogeneity) to determine which one eats the response. Each is a measurement this system can support, because every variable in it is observable.

5.4 Limits

One model, one parameterisation, thirty-two runs. Collective identity is operational (membership overlap between samples), and the fission detector has thresholds, though the results survive varying its floor. The port is distributionally, not trajectory, faithful to the original implementation. The deepest caveat is stated in the Results: generation and time cannot be fully deconfounded in observational data from growing populations, which is why the response claim is stated negatively. Finally, because the grouping subsidy is itself density-dependent, part of the collective-level selection differential could in principle be carried by each agent's own neighbour count; separating the two requires a contextual analysis that regresses agent reproduction on own density and collective size jointly, which I leave to future work. These are limits on

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scope, not on the central measurement: a positive, controlled, compounding selection differential at a level of organisation no line of the model refers to.

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Declarations

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Competing interests. The author declares no competing interests.

Ethics approval and consent to participate. Not applicable.

Data and code availability. The vectorised model, the collective-genealogy tracker, every analysis script, and the stored census behind each figure are openly available at https://github.com/drnamlabs-gif/vitanova.

Use of generative AI. During the preparation of this work the author used generative AI tools to draft and edit the text. After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

Author contributions. N.H.L. designed the study, wrote the code, performed the analyses and wrote the manuscript.

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